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Application of configuration principle on knowledge-based engineering for manufacturing system design

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Abstract

The design of manufacturing systems is a knowledge-intensive activity in which designers from various disciplines are involved, and the lack of effective support for the multi-disciplinary knowledge interaction among designers is one of the most critical reasons for the design failure. The authors intend to combine the configuration design principle with knowledge-based engineering to support the multi-disciplinary knowledge interaction during the intelligent design process. The paper firstly conducts an in-depth review of current intelligent design approaches. Then, the research gap in achieving multi-disciplinary knowledge interaction is pointed out. Finally, the multi-disciplinary knowledge configuration design approach is proposed.

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Keywords: Intelligent design; Multi-disciplinary knowledge interaction; Knowledge configuration; Complex product design

1. Introduction

Applying design knowledge in the design process and realising intelligent design can greatly avoid repetitive design work, save development costs and improve development efficiency [1]. However, the design of industrial manufacturing systems is a knowledge-intensive activity in which designers from various disciplines are involved, and the lack of effective support for the multi-disciplinary knowledge interaction among designers is one of the most critical reasons for the design failure [2].

Currently, the knowledge interaction among designers of different disciplines is always achieved by integrating various existing knowledge bases [3], which is not only a time-consuming task, but might cause a high degree of knowledge redundancy as well. As a result, such knowledge interaction approach can hardly provide a quick response when faced with the market changes or customers' preferences.

In this paper, the authors apply the configuration design principle on knowledge-based engineering to support the multi-disciplinary knowledge interaction during the intelligent design process of manufacturing systems. Based on the configuration design principle, the existing multi-disciplinary knowledge bases can be decomposed and reorganised into different knowledge components, and the new knowledge base which can be used for the design of the required manufacturing system can be constructed by configuring the knowledge components. Compared to the traditional knowledge base integration approach, the proposed knowledge configuration approach can greatly improve the knowledge interaction efficiency of design knowledge.

The structure of this paper is organised as follows: Section 2 conducts a deep review on current intelligent design approaches and presents the existing knowledge gap in achieving multi-disciplinary knowledge interaction in current intelligent design approaches is pointed out. Section 3 introduces the multi-

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disciplinary knowledge configuration approach. An industrial design case in which the proposed multi-disciplinary knowledge configuration approach can be implemented is presented in Section 4. Section 5 draws the conclusion and points out the future research.

2. Literature review

2.1. Knowledge-based engineering approaches

Engineering design is considered as a knowledge-based creative activity in which the designers follow certain processes to design or improve products based on their engineering knowledge [4]. Against this background, the concept of KBE is proposed, and a number of similar definitions of KBE have been put forward by different researchers. Cooper and La Rocca propose one of the definition which has been widely accepted, who defined KBE as a design approach "to solve the design problems by capturing and reusing product and process engineering knowledge in a convenient and maintainable fashion" [5]. Compared to the traditional CAD-based approaches that only focus on the "what" problem of products (i.e., physical characteristics relating to dimensions, geometric and surface contours, and materials), KBE-based design approaches consider not only the abovementioned "what" problem but the problems of "how" (i.e., design process) and "why" (i.e., design rules and regulations) as well [6]. Therefore, the application of KBE approach in the design of manufacturing systems can greatly help designers to avoid repeated development work, improve development efficiency and save development costs, thus attracting attention from both academia and industry. Sadek et al. adopt KBE approach in the conceptual design stage of the industrial product-service system [7]. La Rocca et al. propose a KBE approach to solve the problems relating to aircraft design and optimisation [8]. Yang et al. apply the design knowledge on the structure design of the hull [9]. The above review on current KBE approaches indicates that the KBE approach can be used as effective support for the discipline-specific design. However, the design of manufacturing systems is an activity in which designers from different disciplines are involved, and a discipline-specific KBE approach cannot fully support the design of manufacturing systems.

2.2. Multi-disciplinary knowledge interactions

Traditionally, the interactions between designers from different disciplines can be achieved through meetings, emails or collaborative design platforms. Nowadays, the integration of design data is the most commonly used means of interaction between designers. Currently, design data are stored in the product model in different forms. Product model includes all the information that can be accessed, stored, served and reused to support the product data management functions of product lifecycle management. Various product modes, such as IGES (Initial Graphics Exchange Specification) [10], STEP (STandard for the Exchange of Product) [11], CPM (Core Product Model) [12] and PPO (Product, Process and Organisation model) [13], are proposed to provide designers with a unified data model to aid them in achieving the data-level interaction. However, the design data are directly generated by different designers during the design process, and they are not given a clear meaning by the designer, which often results in confusion and misunderstanding of designers from different disciplines.

Knowledge is considered as the data that are given meaning. During the design process, designers can generate their own design knowledge by integrating their experiences and personal understandings of design data, which directly reflect the designer's design philosophy and rules, thereby guiding the design activities. At present, designers mainly obtain the design knowledge of other disciplines through querying the different knowledge bases. By using the knowledge base, designers currently adopt two methods to achieve the design knowledge interaction. The first method is to establish the relationships between existing knowledge bases, and designers can actively query the knowledge of other disciplines according to design needs. In this way, Sapuan et al. develop a material knowledge base by using the knowledge base tool named KEE (Knowledge Engineering Environment) for automotive design to help the mechanical engineers who are not familiar with the material technology to inquire and select the materials for mechanical components [14]. Torres et al. establish a product quality knowledge base based on CATIA V5, which enables mechanical engineers to consider product quality requirements when designing mechanical components [15]. However, due to the lack of a unified interaction agreement between different knowledge bases, it is a time-consuming task for designers to understand and learn the knowledge stored in other knowledge bases. The second method to achieve the knowledge interaction is to establish a new multi-discipline knowledge base by considering the design knowledge of all disciplines involved in the design process, which provides designers from different disciplines with design knowledge in a unified structural and semantic expression. Tian et al. develop a knowledge base for drones by integrating the knowledge from the disciplines of control, mechanics, and aerodynamics [16]. Wang et al. collect the knowledge from the mechanical, electrical and control domains to support the mechanical engineers, electrical engineers and control engineers to achieve the rapid design of process chambers of integrated circuits [17]. In the second method, a new knowledge base should be constructed by directly capturing the knowledge of various disciplines involved in the design process, so it is suitable for the design of systems with a low-level of complexity. With the rapidly increasing complexity of manufacturing systems, more and more disciplines are involved in the design process. As a result, constructing the design knowledge base for manufacturing systems by capturing all the involved disciplines requires a large amount of time. Considering the limitations of the two above-mentioned methods, the integration of multi-disciplinary knowledge bases is widely used to help designers to achieve knowledge interactions.

Designers always construct the design knowledge bases of their own disciplines. As a result, constructing a multi-disciplinary knowledge base by integrating the existing discipline-specific knowledge bases has gradually become a mainstream approach to realise the knowledge interaction of

designers from different disciplines. Due to the differences in discipline characteristics, design experiences and even educational backgrounds of designers, the ways the designers represent their knowledge in the knowledge bases are not the same. Considering this problem, the existing research mainly focuses on the structural or semantic integration of knowledge representation. XML (eXtensible Markup Language) is a markable language that defines a set of rules for encoding documents in a format that is both human-readable and machine-readable. As a result, it is always regarded as an effective means to achieve the structural integration between two different knowledge bases. Currently, XML has been used to achieve the data integration among CAD, CAM, CAE and CAPP [18, 19]. Although XML provides designers with a common syntax, the semantic differences in the representation of knowledge by designers from different disciplines severely hinder the integration of knowledge bases. The domain ontology can be used as a potential solution to resolve the semantic differences between different knowledge bases. In knowledge engineering, an ontology is defined as a "formal specification of conceptualization" that can be used to express a set of concepts and their relationships for a particular purpose of a domain of interest [20]. Therefore, the domain ontology provides the concepts and their relationships in the knowledge bases with standardized descriptions, so that the problem of semantic differences can be solved. By using an ontology, researchers have achieved the integration of knowledge bases in different disciplines [21,22]. However, this type of integration between knowledge bases only realizes the knowledge addition, which may cause a large amount of redundancy in the new knowledge base. Moreover, the design knowledge in different disciplines may be different or even conflicting. Therefore, integrating the knowledge base directly without considering the compatibility of knowledge cannot improve the knowledge interactions between designers, but on the contrary, it might sometimes hinder the multi-disciplinary design process.

Considering the limitations of current approaches for multi-disciplinary knowledge interactions, the authors propose a novel design approach that applies the configuration design principle to knowledge-based engineering. The details of the proposed design approach will be presented in the following sections.

3. Multi-disciplinary knowledge configuration design

3.1. Principle of configuration design

Configuration design was first proposed by Freeman and Newell in 1971. Based on the principle of configuration design, designers can complete the design task by selecting and integrating the components with certain functions to meet specific requirements. Subsequently, configuration design was widely used for software development, because it provides software engineers with a development method that uses existing software components to construct new systems, so that the compatibility and quality of the software can be improved and its development cost and lead-time can be decreased. With the introduction of system engineering, the concept of component has been broadened. It can be a unit of a physical system, or even an abstract thing, such as method, step, or organisation. Therefore, applying the principle of configuration design on knowledge-based engineering can provide designers with a new perspective for knowledge interactions during the design process of manufacturing systems.

The main research of configuration design includes:

- Definition of interactive components: the functional characteristics of the components (i.e., the services that the components can provide), the structural characteristics of the components (i.e., the content of the components) and the granular characteristics of the components (i.e., component decomposition)
- Interface characteristic analysis: interface information model, interface compatibility rules and system compatibility analysis
- Configuration research: the achievement of rapid component configuration through interfaces to customize the required system

In recent years, with the continuous studies on the configuration design, researchers have applied the principle the configuration design in various manufacturing domains. Komoto and Tomiyama apply the configuration design principle on the design process of complex systems to support their conceptual design stage [23]. Li et al. use the configuration design principle to realize the user-customized design of industrial robots [24]. Bourne et al. believe that the configuration design principle can be an effective support to help designers to provide better business service to satisfy the customer's business requirement [25]. In this paper, the authors apply the configuration design principle on KBE design approach for manufacturing systems in order to support the knowledge interactions during the design process of manufacturing systems.

3.2. Multi-disciplinary knowledge configuration design approach

The framework of the proposed multi-disciplinary knowledge configuration design approach is illustrated in Fig. 1.

First, the knowledge stored in the knowledge bases of various disciplines should be decomposed and reorganized into different knowledge components. Then, according to the specific requirements of the manufacturing system, the product model of the required manufacturing system is used to select the knowledge components. Finally, the selected knowledge components are integrated to construct a multi-disciplinary knowledge base for the required manufacturing system.

The above-mentioned three steps of the framework can be summarized as (1) knowledge decomposition, (2) knowledge component modelling and (3) configuration of knowledge components. The technical solutions to each step will be presented in detail hereafter.

(1) Knowledge decomposition: during the design process, designers incorporate their own practice, experience and personal understanding into design data to form the design knowledge in their own disciplines. Therefore, constructing the mapping between the design knowledge and design data

between different disciplines, the interactive design knowledge between disciplines can be selected. The technical solutions to achieve the knowledge decomposition can be presented as follows:

- Multi-disciplinary design knowledge extraction: the knowledge in a certain discipline can be obtained by interviews, surveys, inquiries of an existing knowledge base. The ambiguity of the knowledge expression should be studied and reduces. In the same discipline, the expression of design knowledge at the semantic level does not always cause misunderstanding among designers due to the standardisation of vocabulary; however, the designers from different disciplines may adopt various syntax to express the knowledge due to the lack of unified syntax. The XML can be used as an effective support to unify the syntax of data, and it can therefore provide the expressions of multi-disciplinary knowledge with a unified syntax.
- Multi-disciplinary design data extraction: the interface is the functional or physical connection between different components of a system, so the multi-disciplinary design data of the manufacturing system is transmitted and exchanged through the interfaces between different components in the product. The authors' research team have adopted UML to establish the multi-disciplinary interface model [26], which can systematically and comprehensively extract multi-disciplinary design data.
- Establishment of the mapping between the multi-disciplinary knowledge and the multi-disciplinary data: once the proposed multi-disciplinary interface model has been instantiated by using the multi-disciplinary design data, the mapping between the multi-disciplinary interface model and the captured design knowledge can be constructed, and the design knowledge can therefore be decomposed according to the mapping result.

(2) Knowledge component modelling: during the configuration design process, the same function can be achieved by different components, whereas similar or even identical component can realize different functions in different design contexts. Therefore, a common representation for the component is required such that a large and complex system can be configured from components by designers from different disciplines. Therefore, after the knowledge base is decomposed into knowledge component in the previous step, a formal and common representation for the knowledge components should be proposed. The technical solutions of knowledge component modelling are presented hereafter:

- Unification of the structure and semantics of multi-disciplinary knowledge: due to the lack of the same syntax and unified vocabulary standardisation, the designers from different always use different ways to express knowledge. Therefore, the unification of the structure and semantics of multi-disciplinary knowledge is the basis of knowledge modelling. Currently, different tools and methods, such as UML, XML, object-oriented modelling approach, first-order logic-based approach, domain ontology, etc., have been adopted for knowledge modelling. Among them, the ontological model can be used as effective support for knowledge modelling. On the one hand, it can simultaneously solve the problem of inconsistency in the syntax and semantic representations of knowledge. On the other hand, it is an intuitive approach, and designers can easily understand and describe concepts and their relationships with the design knowledge. In addition, the utility of ontology to express knowledge can also realize the reasoning of design knowledge. In other words, the combination of different knowledge cannot only realize the integration of multi-disciplinary knowledge, but also generate new knowledge by using the existing knowledge based on the relationships and reasoning rules between concepts. The proposed design approach adopts the OWL (Web Ontology Language) as the ontology modelling language and the Protégé as the ontology modelling software to construct the ontology of knowledge.
- Knowledge component modelling: in order to achieve consistency in syntactic and semantic level, the knowledge components are still modelled by ontology. Since knowledge components in different disciplines may have different expression granularities, and the interaction between the multi-disciplinary knowledge components with different granularities is meaningless. Therefore, the granularity of knowledge components is also a problem that needs to be studied and solved for modelling the knowledge components. The authors' research team have proposed a decomposition theory for decomposing the complex systems [27], which can help designers to obtain an appropriate granularity for modelling the knowledge components.

Interface modelling between knowledge components: the interface between knowledge components is a channel for multi-disciplinary knowledge interaction. Therefore, the representation of the interfaces between knowledge components in different disciplines must be consistent in semantics and structure. This paper proposes four types of information (i.e., material, energy, data and control) and design-related attributes for each type to describe the details of the interfaces between knowledge components and. This interface modelling process can be also supported by using OWL and Protégé.

(3) Configuration of knowledge components: according to the design requirements of the manufacturing system, designers should complete the configuration of knowledge components, and then build a multi-disciplinary design knowledge base to achieve the interaction of multi-disciplinary knowledge. The technical solutions to achieve the configuration of knowledge components will be presented as follows:

- Compatibility between knowledge components: the design requirements from different disciplines for the system may be different or even conflicting. Therefore, in the process of the multi-disciplinary knowledge configuration step, it is very necessary to check the compatibility between knowledge components. The authors' research team have already proposed interface compatibility rules to check the compatibility between components of complex systems [28]. These interface compatibility rules can be used to eliminate the combinations in which two knowledge components are incompatible with each other.

- Optimisation of knowledge component configuration: the configuration of knowledge components is the process of selecting the most suitable knowledge components and constructing the multi-disciplinary knowledge base. On the one hand, this selection process needs to consider the specific design requirements of the designed manufacturing system. On the other hand, designers should also avoid knowledge redundancy caused by repeated descriptions of the same knowledge in different disciplines, and eliminate the conflicts of design knowledge components to achieve the optimal configuration. Therefore, the optimisation of the knowledge component configuration process can be transformed into an MADM (multi-attribute decision-making) problem. Designers can adopt the MADM approach to achieve the configuration optimisation process.

The effectiveness of the multi-disciplinary knowledge configuration design approach will be demonstrated by an industrial case based on the robotic welding system.

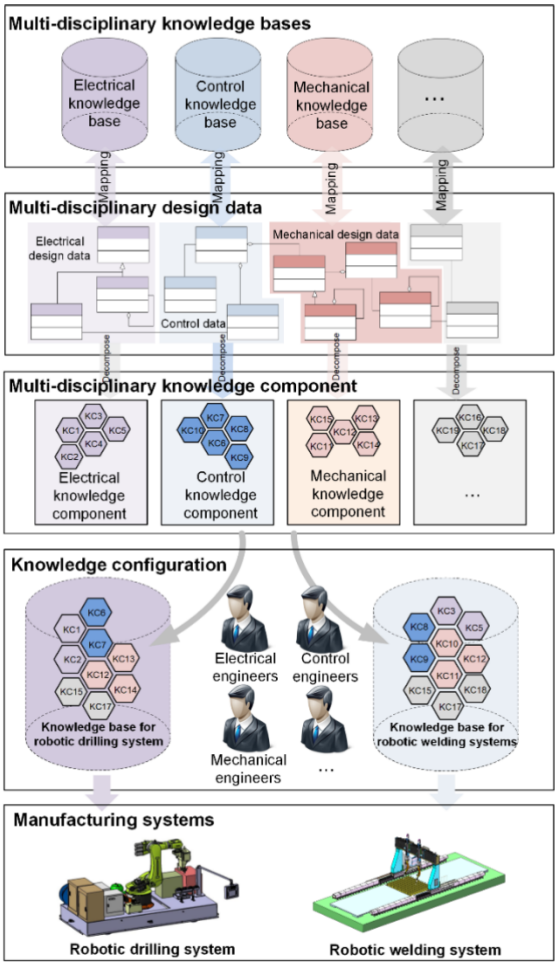


Fig. 1. Framework of multi-disciplinary knowledge configuration design approach

4. Case study

A shipbuilding company requires the authors' design team to design a robotic welding system for robotizing welding tasks.

Fig.2(b) illustrates the architecture of the robotic welding system. The blue boxes represent the main modules, and the purple boxes are the interfaces among modules. The architecture of the robotic welding systems can be summarized as follows: supported by the position detection module (CM^7) which can detect the welding seams, the off-line program sub-system (CM^2) generates the collision-free trajectory. Executing the off-line program, the control module (CM^4) enables the movement module (CM^1) to move the end effector module (SM^1) to the starting point of the welding seams.

Fig. 2 (a) represents the knowledge bases which store the disciplines-specific knowledge of modules. The multi-disciplinary knowledge used for the design of the required robotic welding system can be obtained by mapping the existing discipline-specific knowledge bases and the architecture of the required robotic system and integrated in the multi-disciplinary knowledge base, in which the multi-disciplinary knowledge is represented by the ontology (Fig. 2(c)). The designers from different disciplines can select the specific-discipline components from the multi-disciplinary knowledge base and generate different solutions for the required robotic welding systems. The upper part of Fig. 2(d) illustrates five solutions of the robotic welding systems. After evaluating each solution by using the MADM approach, the most suitable solution can finally be decided.

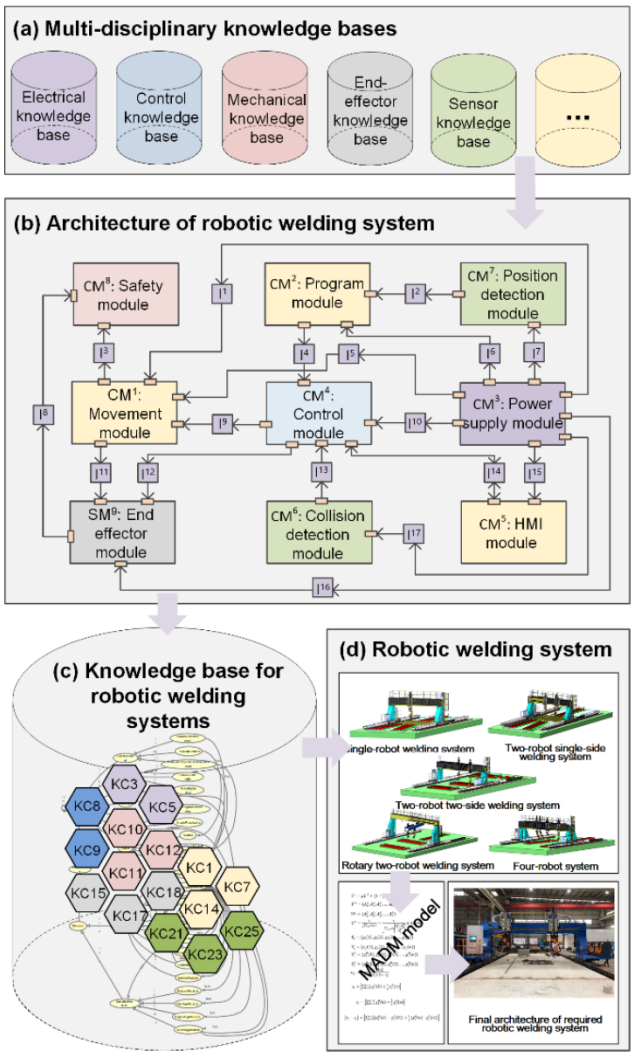


Fig. 2. Implementation of multi-disciplinary knowledge configuration design approach

5. Discussion and conclusion

This paper proposed a knowledge configuration design approach for manufacturing systems to support the multi-disciplinary knowledge interaction, which combines the configuration design principle with knowledge-based engineering. Compared to the proposed configuration design approach, we present the limitations of current approaches for multi-disciplinary knowledge interaction in Table 1.

Table 1. Limitations of current approaches for multi-disciplinary knowledge interactions

Knowledge interaction approach		Limitations
Discipline-specific KBE approaches [7-9]		They cannot support the multi-disciplinary knowledge interactions.
Product model-enabling approaches [10-13]		They only achieve the data-level interactions.
Knowledge base-enabling approaches	Knowledge querying approaches [14,15]	It is a time-consuming task to query the knowledge from other knowledge bases or construct a multi-disciplinary knowledge base.
	Knowledge capturing approaches [16-17]	
	XML-based knowledge integration approaches [18, 19]	They only achieve the structural level knowledge integration, and may cause knowledge redundancy.
	Ontology-based knowledge integration approaches [21, 22]	They do not consider the compatibility of knowledge.

Based on the configuration design principle, the existing multi-disciplinary knowledge bases can be decomposed and reorganised into different knowledge components, and the new knowledge base used for the design of the required manufacturing system can be constructed by configuring the knowledge components. In the future, it is necessary to develop an IT platform or software to manage the multi-disciplinary knowledge components, thereby ensuring access exclusively to authorised designers and avoiding malicious ones.

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References

[1] Chapman CB, Pinfold M. The application of a knowledge based engineering approach to the rapid design and analysis of an automotive structure. Adv. Eng. Softw. 2001;32(12):903–12.
[2] Gulla J. Seven reasons why information technology projects fail. In: SHARE Conference. Orlando, USA.; 2011.
[3] Neches R, Richard F, Tim F. Enabling technology for knowledge sharing. AI Mag. 1991;12(3):36–56.
[4] Shetty D, Kolk RA. Mechatronics system design. Boston, USA: Cengage Learning; 2010.
[5] Cooper D, La Rocca G. Knowledge-based techniques for developing engineering applications in the 21st century. In: 7th AIAA Aviation Technology, Integration and Operations Conference. Belfast, Northern Ireland; 2007.
[6] Mohammed J, May J, Alavi A. Application of computer aided design (CAD) in knowledge based engineering. In: Proceedings of the 2008 IAJC-IJME international conference. Nashville, USA; 2008.
[7] Sadek T, Theiss R. Knowledge based assistance for conceptual development of industrial Product-Service Systems. Adv Intell Soft Comput. 2010;66:1647–64.
[8] La Rocca G, van Tooren MJ. Knowledge-based engineering to support aircraft multidisciplinary design and optimization. J Aerosp Eng. 2010;224(9):1041–55.
[9] Yang HZ, Chen JF, Ma N, Wang DY. Implementation of knowledge-based engineering methodology in ship structural design. Comput Des. 2012;44(3):196–202.
[10] Kemmerer SJ. STEP: The grand experience. Gaithersburg, USA: US Department of Commerce, Technology Administration, National Institute of Standards and Technology; 1999.
[11] ISO10303-1. Overview and fundamental principles. ISO; 1994.
[12] Fenves S, Foufou S, Bock C, Sriram RD. CPM : a core model for product data. J Comput Inf Sci Eng. 2006;8(1):1–14.
[13] Robin V, Girard P. An integrated product-process-organisation model to manage design system. In: IMACS Multiconference on Computational Engineering in Systems Applications. Beijing, China. 4-6 October; 2006.
[14] Sapuan SM. A knowledge-based system for materials selection in mechanical engineering design. Mater Des. 2001;22(8):687–95.
[15] Torres VH, Ríos J, Vizán A, Pérez JM. Approach to integrate product conceptual design information into a computer-aided design system. Concurr Eng Res Appl. 2013;21(1):27–38.
[16] Tian F, Voskuijl M. Automated generation of multiphysics simulation models to support multidisciplinary design optimization. Adv Eng Informatics. 2015;29(4):1110–25.
[17] Wang KS, Lin J, Cheng J, Ji LH. Investigation on the development of knowledge-based engineering and its application in rapid design of process chamber of IC equipment. Appl Mech Mater. 2013 Aug;373:2147–55.
[18] Zhang Y, Li H. A method of CAD/CAE data integration based on XML. In: International Conference on Computer Science & Education (ICCSE). Singapore, Singapore; 2011.
[19] Šormaz D, Arumugam J, Harihara R, Patel C, Neerukonda N. Integration of product design, process planning, scheduling, and FMS control using XML data representation. Robot Comput Integr Manuf. 2010;26:583–95.
[20] Gruber T. A translation approach to portable ontology specifications. Knowl Acquis. 1993;5(2):199–220.
[21] Vegetti M, Leone H, Henning G. PRONTO: An ontology for comprehensive and consistent representation of product information. Eng Appl Artif Intell. 2011 Dec;24(8):1305–27.
[22] Maree M, Belkhatir M. Addressing semantic heterogeneity through multiple knowledge base assisted merging of domain-specific ontologies. Knowledge-Based Syst. 2015;73(1):199–211.
[23] Komoto H, Tomiyama T. A framework for computer-aided conceptual design and its application to system architecting of mechatronics products. Comput Des. 2012;44(10):931–46.
[24] Li J, Daaboul J, Tong S, Bosch-Mauchand M, Eynard B. A design pattern for industrial robot: User-customized configuration engineering. Robot Comput Integr Manuf. 2015;31(C):30–9.
[25] Bourne S, Szabo C, Sheng Q. Transactional behavior verification in business process as a service configuration. IEEE Trans Serv Comput. 2017;2(99):1–14.
[26] Zheng C, Le Duigou J, Bricogne M, Eynard B. Multidisciplinary interface model for design of mechatronic systems. Comput Ind. 2016 ;76(2):24–37.
[27] Zheng C, Le Duigou J, Bricogne M, Dupont E, Eynard B. Interface model enabling decomposition method for architecture definition of mechatronic systems. Mechatronics. 2016;40:194–207.
[28] Zheng C, Qin X, Eynard B, Li J, Bai J, Zhang Y, et al. Interface model-based configuration design of mechatronic systems for industrial manufacturing applications. Robot Comput Integr Manuf. 2019;59:373–84.